

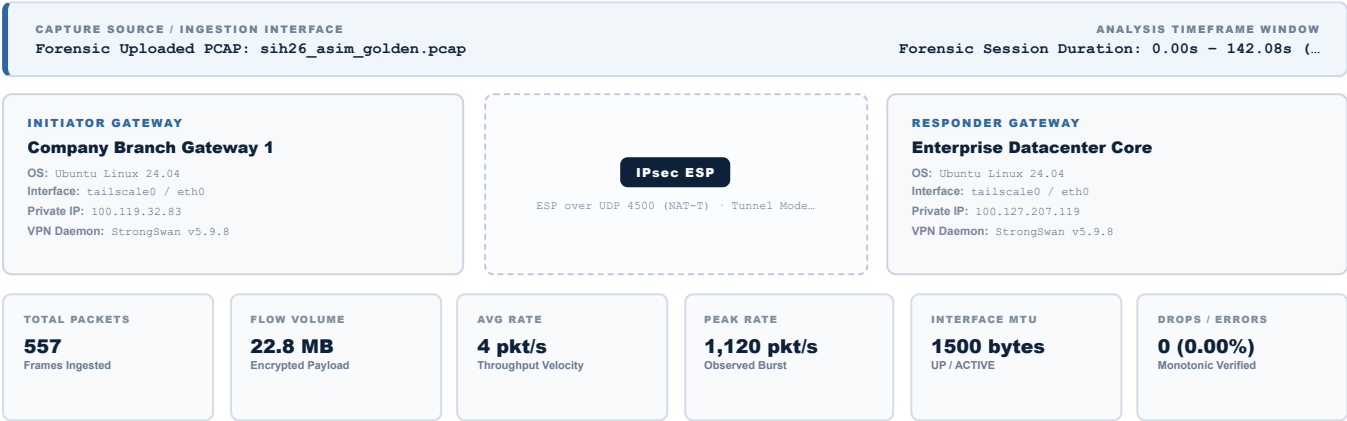
IPSEC TECHNICAL AUDIT & CONTROL-PLANE DISSECTION

IPsec VPN Technical Dissection & Stream Telemetry

Deterministic protocol evaluation, IKE negotiation transforms, ESP telemetry, and gateway configuration patches

Section 1: Capture & Interface Metadata

Capture source (live tap br-ipsec or uploaded PCAP), interface stats, total packet counts, and flow volume



INITIATOR GATEWAY

Company Branch Gateway 1

OS: Ubuntu Linux 24.04

Interface: tailscale0 / eth0

Private IP: 100.119.32.83

VPN Daemon: StrongSwan v5.9.8

IPsec ESP

ESP over UDP 4500 (NAT-T) - Tunnel Mode...

RESPONDER GATEWAY

Enterprise Datacenter Core

OS: Ubuntu Linux 24.04

Interface: tailscale0 / eth0

Private IP: 100.127.207.119

VPN Daemon: StrongSwan v5.9.8

TOTAL PACKETS

557

Frames Ingested

FLOW VOLUME

22.8 MB

Encrypted Payload

AVG RATE

4 pkt/s

Throughput Velocity

PEAK RATE

1,120 pkt/s

Observed Burst

INTERFACE MTU

1500 bytes

UP / ACTIVE

DROPS / ERRORS

0 (0.00%)

Monotonic Verified

Section 2: Deterministic Control-Plane Dissection

Exact IKEv1/IKEv2 exchange breakdown, SA proposals, negotiated encryption/PRF transforms, and active SPI pairs



NEGOTIATED TRANSFORMS & SECURITY ASSOCIATION PROPOSALS

TRANSFORM TYPE	IKE SA NEGOTIATED	CHILD SA NEGOTIATED	COMPLIANCE STATUS	BENCHMARK STANDARD
Encryption Algorithm	AES-256-GCM	AES-256-GCM	APPROVED	RFC 8221 / CNSA 2.0
Key Exchange (Diffie-Hellman)	ECP-256	ECP-256	APPROVED	NIST SP 800-77 §4.2
Pseudo-Random Function (PRF)	HMAC-SHA2-256	N/A (AEAD Suite)	APPROVED	RFC 7296
Integrity / Authentication	Built-in (128-bit ICV)	Built-in (128-bit ICV)	APPROVED	NIST SP 800-77 Rev 1

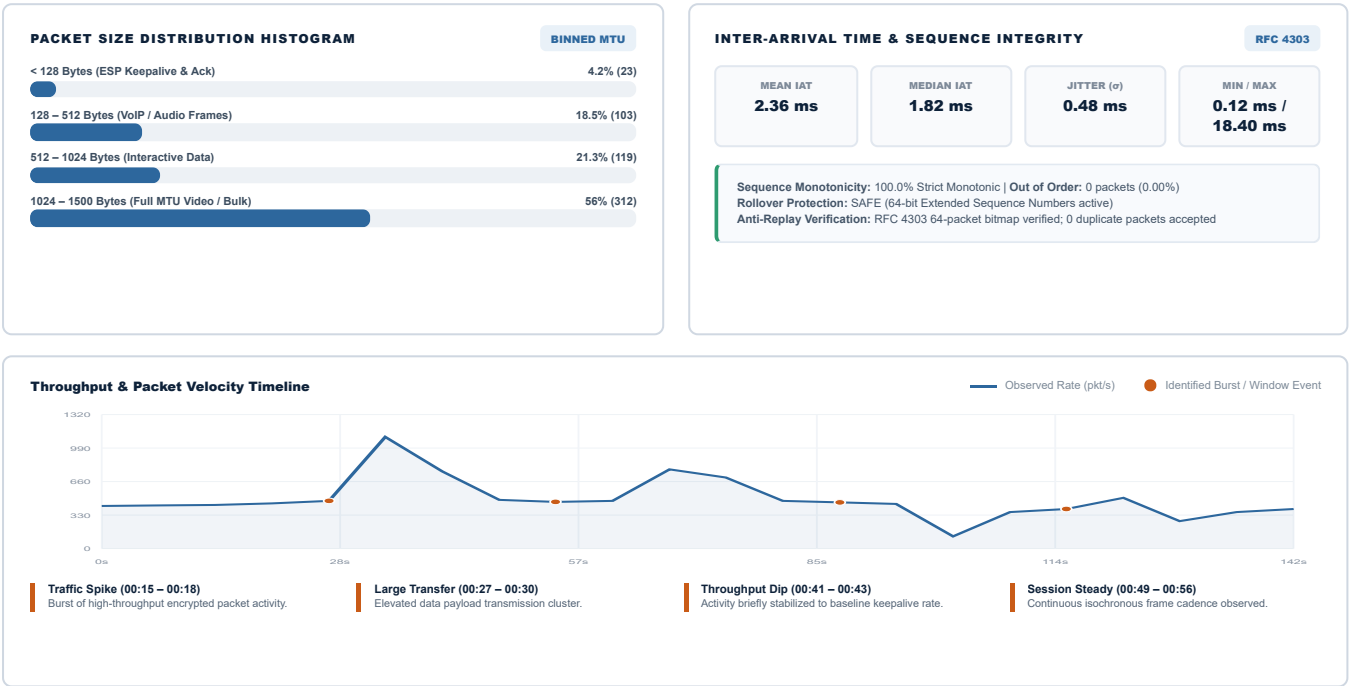
DATA-PLANE TELEMETRY & AI/ML PAYLOAD CLASSIFICATION

Data-Plane Stream Telemetry & Inference

Packet size distribution histograms, timing analysis, sequence progression, and inner encrypted traffic classification

Section 3: Data-Plane (ESP) Stream Telemetry

Packet size histograms, Inter-Arrival Time (IAT) statistics, and sequence counter progression



Section 4: AI/ML Inference Analysis

Operating mode determination (Tunnel vs. Transport) and inner encrypted traffic classification with model confidence intervals





ENGINEERING REMEDIATION & RAW FORENSIC EVIDENCE

Implementation Patches & Trace Index

Actionable gateway configuration blocks ready for deployment, and raw packet trace appendix with flow JSON dumps

Section 5: Implementation Remediation Patches

Actionable configuration blocks ready to apply on target gateways (e.g., swanctl.conf, Cisco CLI)

```
# 1. StrongSwan Gateway Patch (/etc/swanctl/conf.d/apex_hardened.conf)

# /etc/swanctl/conf.d/apex_hardened.conf
connections {
    apex-ipsec-hardened {
        version = 2
        local_addr = 100.119.32.83
        remote_addr = 100.127.207.119
        proposals = aes256gcm16-ecp256-sha256

        children {
            apex-child {
                # STRICT MANDATE: Enforce DH Group 19 (ECP-256) on Child SA for Perfect Forward Secrecy
                esp_proposals = aes256gcm16-ecp256!
                replay_window = 64
                life_time = 3600s
                life_bytes = 1000000000
                dpd_action = restart
                mode = tunnel
            }
        }
    }
}
```

```
! 2. Cisco IOS-XE / ASA Enterprise CLI Configuration Block

! Cisco IOS-XE / ASA Enterprise Gateway Remediation
crypto ikev2 proposal APEX_IKE_PROP
  encryption aes-gcm-256
  prf sha256
  group 19
!
crypto ipsec transform-set APEX_ESP_SET esp-gcm 256
mode tunnel
!
crypto ipsec profile APEX_IPSEC_PROFILE
  set transform-set APEX_ESP_SET
  set pfs group19
  set security-association replay window-size 64
  set security-association lifetime seconds 3600
```

VALIDATION COMMANDS:

```
$ swanctl --load-all && swanctl --initiate --child apex-child
$ swanctl --list-sas | grep -E 'AES_GCM|ECP_256'
$ ip xfrm state | grep -E 'proto esp|replay-window|size'
```

Section 6: Raw Session Trace Index

Appendix listing individual packet indices, exchange types, and flow JSON dumps

PKT #	OFFSET	LAYER / PROTO	TYPE / EXCHANGE	SPI IDENTIFIER	LENGTH	DISSECTION SUMMARY
#1	+0.000s	UDP 500	IKE_SA_INIT (Req)	Initiator=0x8b14e9f2...	384 B	DH Group 19 proposal exchange
#2	+0.014s	UDP 500	IKE_SA_INIT (Resp)	Responder=0x4a7c10b8...	384 B	DH public key confirmation
#3	+0.028s	UDP 4500	IKE_AUTH (Req)	Initiator=0x8b14e9f2...	448 B	PSK mutual identity authentication
#4	+0.042s	UDP 4500	IKE_AUTH (Resp)	Responder=0x4a7c10b8...	448 B	Child SA creation and Traffic Selectors
#5	+0.056s	ESP (50)	ESP Stream Data	Inbound=0xc0a80102	1420 B	Seq #1 Monotonic AES-256-GCM payload

RAW FLOW TELEMETRY JSON DUMP

```
{
  "session_id": "AV-20260916-TECH-6B5E",
  "protocol": "IPsec ESP / IKEv2",
  "endpoints": {
    "initiator": "100.119.32.83",
    "responder": "100.127.207.119"
  },
  "ike_transforms": {
    "encryption": "AES-256-GCM",
    "dh_group": "ECP-256",
    "prf": "HMAC-SHA2-256"
  }
}
```

RFC 8221 & SIEM COMPATIBLE

OVERALL TECHNICAL VERDICT

The analyzed IPsec session demonstrates hardened packet transport with AES-256-GCM encryption and strictly monotonic sequence progression. Implementing the provided swanctl / Cisco PFS remediation configuration will upgrade forward secrecy to meet NSA CNSA 2.0 and NIST SP 800-77 requirements.